

Superheat and Subcooling

Reading what the refrigerant is telling you, and finding the faults a subcooling check alone will miss.

In this handout

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2. Measuring superheat and subcooling
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The one idea to take away

Subcooling tells you about the **liquid side**: is there a solid column of liquid at the metering device? Superheat tells you about the **vapor side**: did the evaporator finish boiling, and is the compressor getting dry gas? A system can pass one and fail the other. If you never put a clamp on the suction line, half the system is invisible to you.

1 · Saturation: the reference point for everything

Inside a coil where liquid and vapor are both present, refrigerant sits at its **saturation temperature**, fixed by its pressure. Read the pressure, look it up on the PT chart, and you know the temperature of the boiling or condensing refrigerant without touching it. Every superheat and subcooling reading is a comparison against one of these two saturation temperatures.

Vsat: vapor saturation temperature

The saturation temperature on the **low side**, from **suction pressure**. It is the temperature the refrigerant boils at in the evaporator. Sometimes called the evaporator saturation temperature or the dew point.

Vsat = PT chart(suction pressure)

Use the **dew** column for blends. R-410A has almost no glide (about 0.2 °F), so dew and bubble are effectively the same; R-32 is a single compound with none. Blends like R-454B glide about 1.5 °F, so the column matters there.

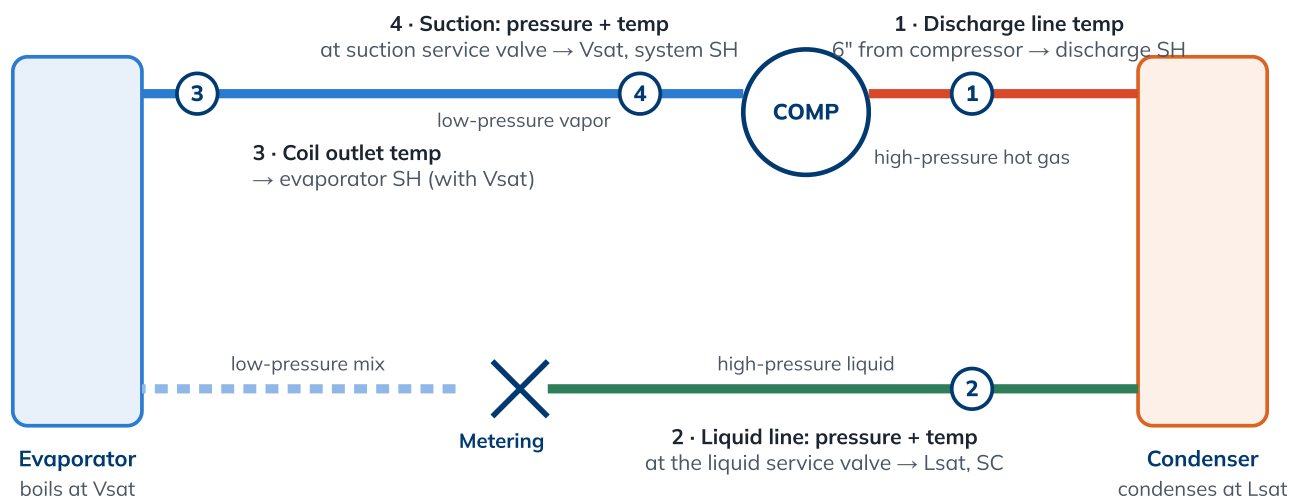
Lsat: liquid saturation temperature

The saturation temperature on the **high side**, from **liquid line or head pressure**. It is the temperature the refrigerant condenses at in the condenser. Also called the condensing temperature or bubble point.

Lsat = PT chart(liquid / head pressure)

Use the **bubble** column for blends. On a heat pump in cooling, the liquid service valve is on the high side; in heating it is too, but the condenser is now the indoor coil.

Where the refrigerant is, and what you can measure there



Pressure gives you saturation. A temperature at the same point tells you how far the refrigerant is from it.

2 · The three numbers, and what each one proves

System superheat (suction superheat, total superheat)

$$\text{System SH} = \text{suction line temp} - \text{Vsatsat}$$

Measure: suction pressure at the suction service valve; suction line temperature 4–6" from the valve with a clean, insulated clamp. Let the system run 15 minutes at steady load first; on an inverter, lock the compressor speed (service / charge mode) or the number will chase the speed.

Proves: how much heat the vapor picked up after it finished boiling. Positive superheat means the gas reaching the compressor is dry. It includes the evaporator's own superheat *plus* any heat the suction line picked up on the way back.

Typical: TXV / EEV systems 5–15 °F at the outdoor unit. Fixed orifice: from the manufacturer's superheat chart (indoor wet bulb and outdoor dry bulb). Inverters: compare with the board's own target (D-Checker logs it).

Evaporator superheat

$$\text{Evaporator SH} = \text{coil outlet temp} - \text{Vsatsat}$$

The same idea measured at the coil outlet. A TXV or the indoor EEV controls this one. When system SH and evaporator SH are far apart, the suction line is the story (heat gain in an attic, a leaking reversing valve), not the metering device.

Subcooling

$$\text{SC} = \text{Lsatsat} - \text{liquid line temp}$$

Measure: liquid pressure and liquid line temperature at the liquid service valve. **Proves:** how far below its condensing temperature the liquid has been cooled, i.e. that a solid column of liquid is leaving the condenser.

Typical: the nameplate / installation-manual target (often 8–15 °F). On TXV and EEV systems, charge to subcooling.

Discharge superheat

$$\text{Discharge SH} = \text{discharge line temp} - \text{Lsatsat}$$

Measure: discharge line temperature 6" from the compressor; Lsatsat from head pressure. **Proves:** what the compressor is doing to the gas, the compressor's own health line. Compression adds heat, and a suction-cooled motor adds more, so dry gas always leaves well above Lsatsat.

- **Under about 25 °F:** wet compression. Liquid is reaching the compressor even if system SH looks acceptable; oil is being washed off the bearings.
- **About 40–80 °F:** the normal band for a residential split at full load.
- **Over about 90 °F, or a discharge line over 225 °F (R-410A):** the compressor is starving or overheating; oil starts breaking down.

Why both: system SH says what is *entering* the compressor; discharge SH says what *happened inside it*. A leaking reversing valve or hot-gas bypass can make suction SH look fine while discharge SH screams.

3 · PT tables: R-410A and R-32

Gauge pressure (psig) to saturation temperature (°F). R-410A is the dew point; R-32 is a pure refrigerant. Between rows, interpolate. Values from CoolProp 8, the same tables the Cycle Viewer uses.

Low side: find Vsat

°F	R-410A psig	R-32 psig
20	78.4	80.0
25	87.4	89.2
30	97.0	99.1
35	107.3	109.7
40	118.4	121.0
45	130.1	133.0
50	142.6	145.8
55	156.0	159.5
60	170.1	174.0

High side: find Lsat

°F	R-410A psig	R-32 psig
85	254.6	260.9
90	274.5	281.3
95	295.4	302.9
100	317.6	325.7
105	340.8	349.6
110	365.4	374.9
115	391.2	401.4
120	418.3	429.3
125	446.8	458.7

Worked example (R-410A, cooling)

Suction: 118 psig, suction line 52 °F.

Vsat = 40 °F → **System SH** = 52 – 40 = 12 °F

Liquid: 341 psig, liquid line 95 °F.

Lsat = 105 °F → **SC** = 105 – 95 = 10 °F

Discharge line: 168 °F.

Discharge SH = 168 – 105 = 63 °F

All three in their normal bands: dry gas into the compressor, solid liquid at the metering device, a compressor running at a healthy temperature.

Do not mix refrigerants. At the same pressure, R-32 saturates 1–3 °F colder than R-410A in this range. Reading an R-32 unit on an R-410A chart makes superheat look about 2 °F higher and subcooling about 2 °F lower than they are. Check the nameplate.

4 · High and low superheat: causes, effects, fixes

High superheat: the evaporator is starving

What it does: less refrigerant boiling means less capacity; the compressor runs hot because the suction gas that cools its motor is thin and warm; discharge temperature climbs and oil breaks down.

Cause	Tell-tale	Fix
Undercharge / leak	SC low, suction low	Find the leak, weigh in
Liquid-line restriction (drier, kink, strainer)	SC high , temperature drop across the restriction	Replace drier / repair line, not charge
TXV underfeeding (lost bulb charge, plugged screen)	SC normal-high, valve cold, no drop at drier	Replace powerhead / valve, clean screen
EEV not opening, or its sensor reading low	Pulses move, SH does not	Valve coil / body, or the thermistor
Suction line heat gain	Coil outlet SH normal, system SH high	Insulate the suction line
High indoor load (fixed orifice)	Return air hot / wet	Normal; recheck at design conditions

Low superheat: liquid is coming back

What it does: liquid reaching the compressor dilutes the oil, washes the bearings and can slug the valves. The house may cool fine while the compressor dies slowly, with no error code.

Cause	Tell-tale	Fix
Low indoor airflow (filter, coil, blower, ducts)	Low V _{sat} , small air ΔT	Restore airflow first
Overcharge (fixed orifice)	SC high, head high	Recover to the chart
Wrong / oversized piston	SC normal, SH near zero	Install the matched piston
TXV bulb loose, uninsulated or misplaced	SC normal, valve overfeeds	Re-strap at 4 or 8 o'clock, insulate
EEV stuck open, or sensor reading high	Valve pinned open	Valve, or the thermistor driving it
Low load (cool / mild day)	Short runs, cool return	Judge at load, check sizing

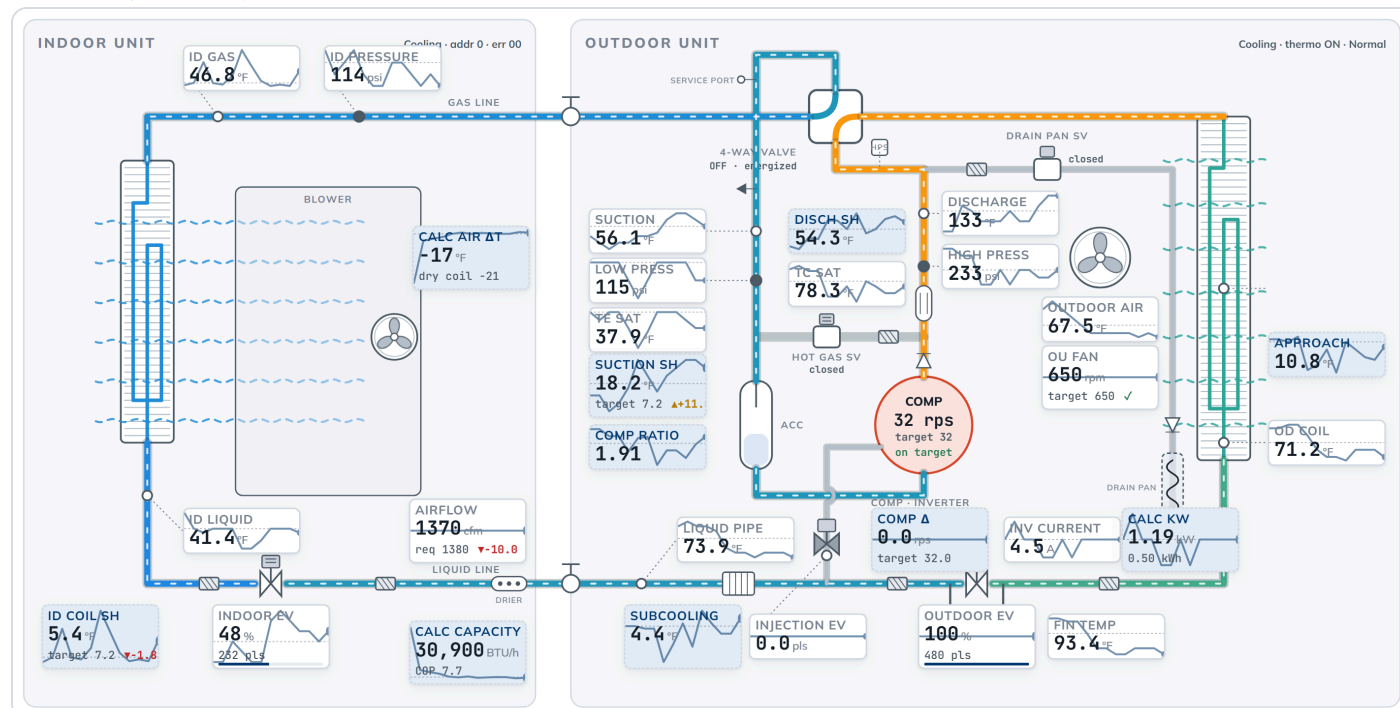
The diagnosis matrix: put superheat and subcooling together

	SC low	SC normal	SC high
SH high	Undercharge leak	TXV/EEV underfeeding or suction heat gain	Restriction drier, line, screen
SH normal	Check charge vs load	Charge OK confirm with discharge SH	Slight overcharge, or condenser airflow
SH low	Low load / airflow with low charge	Airflow, piston, TXV bulb or EEV the row a SC-only check misses	Overcharge fixed orifice

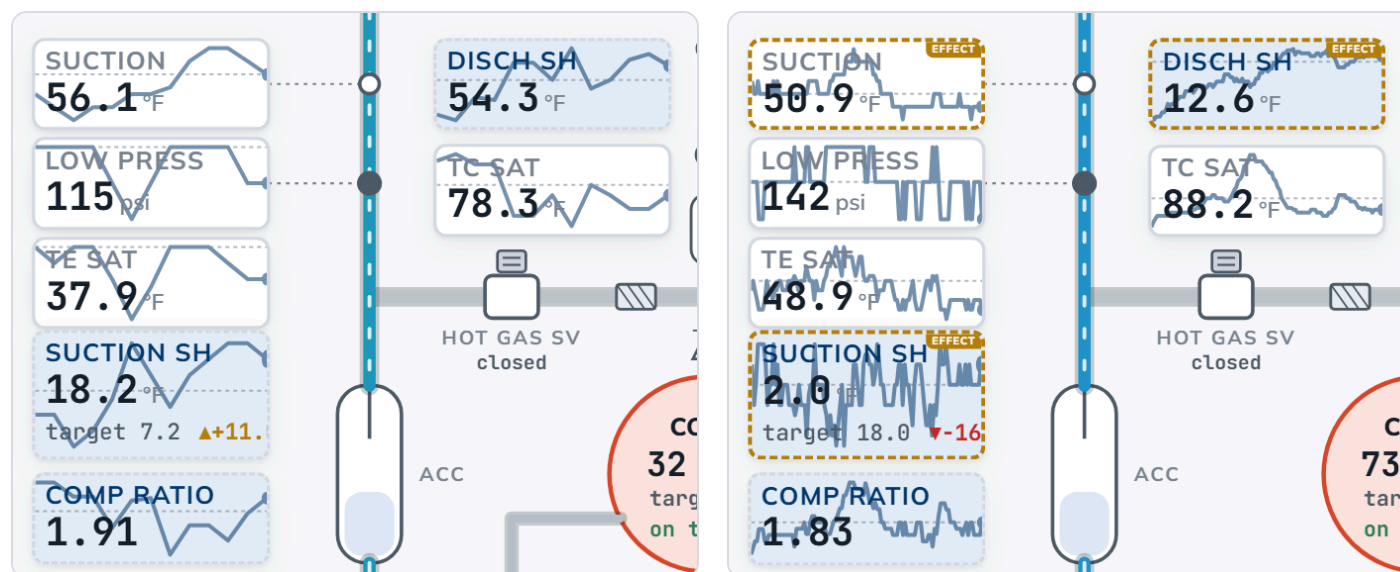
5 • Reading it in the D-Checker Cycle Viewer

On a Daikin inverter, the D-Checker log already has the pressures and temperatures; the Cycle Viewer works out Vs_{sat} (labelled **Te sat**), Ls_{sat} (**Tc sat**), suction SH, discharge SH and subcooling on every row. Tags marked **MEASURED** are the unit's sensors; tags marked **CALCULATED** are worked out from them (hover, or long-press on a phone, to see the formula).

A healthy cooling run



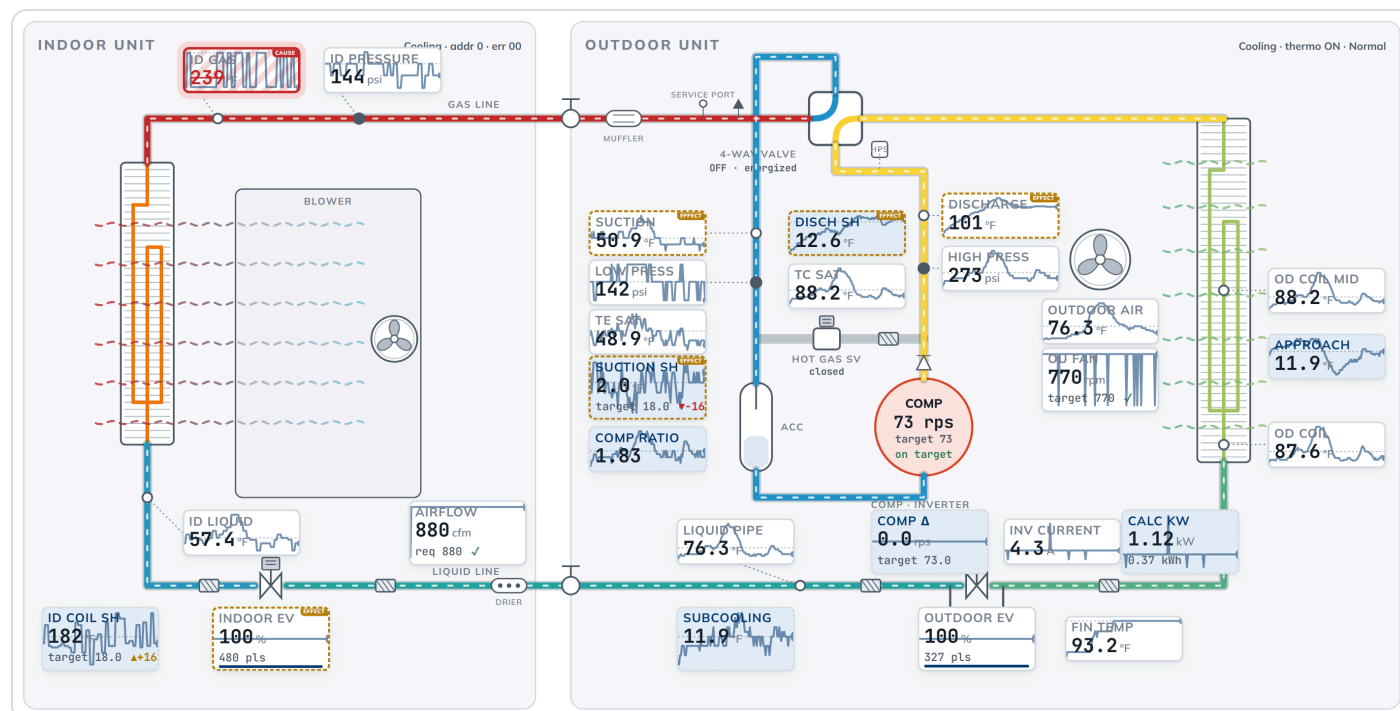
R-32 heat pump in cooling at low speed (32 rps). Suction SH 18 °F and discharge SH 54 °F: dry gas into the compressor, and a compressor running in its normal band.



Dry: Te sat 37.9, suction 56.1 → SH 18.2. Tc sat 78.3, discharge 133 → discharge SH about 54.

Flooding: Te sat 48.9, suction 50.9 → SH 2.0 against an 18 target. Tc sat 88.2, discharge 101 → discharge SH about 13. Liquid is reaching the compressor.

5 · continued: when subcooling looks fine



R-32 heat pump in cooling at full speed (73 rps). **Subcooling 11.9 °F: right where it belongs.** A subcooling-only check calls this charge good and leaves. But suction SH is 2.0 °F, discharge SH 12.6 °F, and the indoor EEV is pinned at 100 %. The viewer marks the cause (ID GAS, reading an impossible 239 °F) and the readings it is driving (EFFECT).

Indoor gas thermistor failed → indoor EEV driven wide open → liquid flooding the compressor

Confirmed by real repair

Evidence: The indoor gas-pipe thermistor read 239°F for 100% of the cooling run, against 58°F on the liquid pipe beside it: a fixed, impossible value, the way an open circuit or an unplugged sensor reads. The indoor board controls its EEV on coil superheat (gas – liquid), so it saw about 182°F of superheat against a 18°F target and did the only thing its logic allows: opened the valve to 100% and held it there at every speed. The coil was overfed, suction superheat sat at 1.8°F and discharge superheat at 11.4°F, so liquid is reaching the compressor. Neither board raised a code: to the indoor board a hot gas pipe is not a fault, it is a reason to open the valve.

1. Resistance-check the indoor gas-pipe thermistor against the service-manual table (it should read near the coil outlet, 40s–50s °F in cooling) and reseal its plug on the indoor board.
2. After the repair, watch the indoor EEV come off 100% and suction superheat climb to its target within a few minutes.
3. Expect a sweating or frosted suction line and a cold compressor shell on site; check the oil if it has run like this for long.

The synopsis explains the cascade. A failed indoor gas-pipe thermistor made the indoor board think the coil was starving, so it opened its valve all the way and flooded the compressor. No error code was raised. Confirmed on the repair: replacing the thermistor fixed it.

6 • Scenario worksheet

Each system below passes a subcooling check. For each: work out Vsat, Lsat, system SH and discharge SH, decide what is wrong, and write the one repair that fixes it. R-410A unless stated.

A • "It cools, but it's loud on start-up"

Attic air handler with a TXV, 3-ton. Customer hears the outdoor unit "rattle" for the first minute. Filter new, coil clean.

OAT 92 °F • return 76 / 63 WB • supply 61
Suction 130 psig • suction line 47 °F
Liquid 341 psig • liquid line 95 °F
Nameplate SC 10 °F

Discharge line 125 °F
Insulation on the suction line ends 1 ft before the TXV bulb

Vsat ____ Lsat ____ System SH ____ Discharge SH ____

B • "Coil replaced last spring, now the compressor is noisy"

Fixed-orifice system, 2½-ton condenser. Indoor coil was replaced; the new coil came with a factory piston. Charge was set to 8 °F subcooling.

OAT 88 °F • return 75 / 63 WB
Manufacturer SH chart target: 12 °F
Suction 136 psig • suction line 49 °F

Liquid 320 psig • liquid line 92 °F
Discharge line 128 °F
Piston stamped for a 4-ton

Vsat ____ Lsat ____ System SH ____ Discharge SH ____

C • "Not keeping up since the drier was changed"

TXV system. A tech changed the liquid-line drier during a repair two weeks ago.

OAT 94 °F
Suction 97 psig • suction line 58 °F
Liquid at condenser 300 psig • 83 °F
Nameplate SC 10 °F

Drier inlet 83 °F • drier outlet 70 °F, sweating
Discharge line 190 °F

Vsat ____ Lsat ____ System SH ____ Discharge SH ____

6 · Scenario worksheet, continued

D · "High superheat, so it needs refrigerant?"

TXV, 65-ft line set run through a 135 °F attic. A previous tech "topped it off" last year.

OAT 98 °F	SC 10
At the condenser: suction 125 psig, suction line 66 °F	Discharge line 190 °F
At the air handler: coil outlet 50 °F	20 ft of suction insulation missing in the attic
Liquid 341 psig · liquid line 95 °F · nameplate	

Vsat ____ System SH ____ Evaporator SH ____ Discharge SH ____

E · "Inverter, no error code, compressor sounds rough" (R-32)

Daikin FIT heat pump in cooling. D-Checker log, compressor locked at 73 rps.

OAT 76 °F · airflow 880 of 880 cfm	liquid pipe 76.3 °F
Low pressure 142 psig · Te sat 48.9 °F · suction 50.9 °F	Discharge 101 °F
Board target SH 18 °F	Indoor EEV 480 / 480 pulses
High pressure 273 psig · Tc sat 88.2 °F ·	Indoor gas pipe 239 °F · indoor liquid pipe 57 °F

System SH ____ SC ____ Discharge SH ____

F · "Heat pump not keeping up in cooling"

Heat pump with a TXV indoor coil, cooling mode. Charge checks out by subcooling.

OAT 95 °F · supply-return ΔT 11 °F	Liquid 318 psig · liquid line 90 °F
Suction 150 psig · suction line at service valve 63 °F	Discharge line 215 °F
	Tube from reversing valve to compressor: 88 °F

Vsat ____ Lsat ____ System SH ____ Discharge SH ____

Quick card: what to carry in your head

- **SC** = Lsat – liquid line. Proves the liquid side.
- **System SH** = suction line – Vsat. Proves dry gas reaches the compressor.
- **Evaporator SH** = coil outlet – Vsat. Separates the coil from the suction line.
- **Discharge SH** = discharge line – Lsat. Proves what happened inside the compressor; under 25 °F is wet, over 90 °F is hot.
- Never charge on one number. Never trust a reading taken while an inverter is changing speed.